

AYUSH MAURYA

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ABOUT ME

CS + Computational Human Sciences dual-degree research student at IIIT Hyderabad, working at the intersection of software systems, ML, and human behavior — from electoral agent simulations to agentic AI platforms.

EDUCATION

IIIT Hyderabad — Dual Degree (B.Tech CSE + M.S. Research) 2021 – Present

Human Sciences Research Centre (HSRC). Advisor: Dr. Aniket Alam.
Research focus: Political Narratives & Electoral Systems in India

Relevant Labs: Design & Analysis of Software Systems · Human Science Lab 2022 – 2024

Mary Gardiner's Convent (XII: 90.1%) · City Montessori School (X: 94.2%)

EXPERIENCE

SERC Digital Twin Environmental IoT

Real-time environmental monitoring dashboard with MERN stack, tracking IoT sensors for temperature, humidity, water, and air quality.

MERN IoT SQL

tafea System Design

Designed student client software system using SDLC methodologies with integrated LLM APIs for natural language interfaces.

System Design UI Design LLM

KEY PROJECTS

ElectoralSim PyPI ↗

Agent-based electoral simulation toolkit modeling 1M+ voter agents with 89× speedup via Polars & Numba across 8+ electoral systems.

Python Agent-Based Modeling Polars Numba

IPL Strategy Lab ipl.ayushmaurya.online ↗

Agentic AI cricket analytics with multi-agent architecture (Analyst → Strategist → Evaluator). Runs Pandas in-browser via Pyodide (WASM) over 260K+ deliveries.

React Agentic AI Pyodide (WASM) Pandas

CrowdTwin crowdtwin.vercel.app ↗

Predictive crowd monitoring dashboard prototype built for Tech Entrepreneurship, featuring event analytics and interactive pitching.

Next.js Product Design Prototype

Design for Social Innovation (TAFEA)

AI-assisted platform helping Teach For India fellows plan and customize extracurricular curricula for resource-constrained classrooms.

Human-Centered Design Education AI

TempusLogic GitHub ↗

NLP diagnostic suite testing LLM logical consistency under adversarial prompting via Dyads/Triads reasoning and GSM8K math stress tests.

Python NLP Research LLM Evaluation

NFS Control Plane

Distributed file-system control plane prototype in C — UDP discovery, per-role TCP connections, multithreaded listeners, and host IP detection.

C Distributed Systems TCP Multithreading

xv6 Kernel Extensions

Extended xv6 OS with custom system calls, timer alarms, process accounting, and copy-on-write fork() using page-fault duplication.

xv6 Operating Systems C

Hoi-shell GitHub ↗

Unix-like shell in C supporting command piping, background execution, history logging/replay/purge, PID inspection, and recursive search.

C Linux Unix Shell Process Mgmt

EXPERTISE

Languages

Python, C/C++, JavaScript/TS, SQL

ML & Data

PyTorch, Agentic AI, Polars, Numba, Pandas, Scikit-learn

Full Stack & Apps

React, Next.js, Flutter, Tailwind, PWA, Pyodide, Bun

Systems

Linux, System Design, Systems Programming

FOR FUN

Music & Sports

Bass guitarist in college band, ukulele, lawn tennis & pickleball

OTHER PROJECTS

ML for Local Geoguesser

ResNet50 location estimation model predicting region, lat/lon, and camera angle with 92% accuracy.

Python PyTorch ResNet50

Age Prediction Model

Transfer learning vs. ensemble methods (Random Forests, SVMs) for age estimation from images.

Machine Learning PyTorch

Highland History Project

OCR & LLM-assisted digitization of Himalayan trade registers with Dr. Aniket Alam at HSRC.

Machine Learning OCR LLM

Campus Mart & Katha

UI/UX designs for buy-sell-rent campus app with NFC tracking and storytelling artisan marketplace.

UI/UX Design Prototyping

Representing India ↗

Computational human sciences research on political narratives & electoral dynamics in India at IIIT-H.

Computational Social Science

WorDrop ↗

Privacy-first offline Android safety app using Vosk speech recognition with configurable hotwords.

Flutter Vosk Android